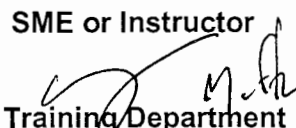
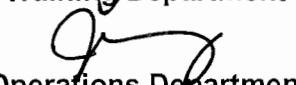


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Conduct of Operations		
TASK:	Calculate Shutdown Margin		
TASK NUMBER:	N1200030301		
JPM NUMBER:	15-01 NRC RO A1-2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.43
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:		Classroom	
REFERENCES:	SC.RE-ST.ZZ-0002(Q), Rev. 23 SHUTDOWN MARGIN CALCULATION S2.RE-RA.ZZ-0016(Q), Rev. 8 CURVE BOOK (both checked 8-9-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	30 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	50 minutes		
Developed By:	G Gauding Instructor	Date:	8-9-16
Validated By:	R Evans / N Mulford SME or Instructor	Date:	9-22-16 / 9-29-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Conduct of Operations

TASK: Calculate Shutdown Margin

TASK NUMBER: N1200030301

INITIAL CONDITIONS:

- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was been declared INOPERABLE at 1000 today due to not moving out during a rod exercise test. You have been directed to perform a Shutdown Margin Calculation IAW SC.RE-ST.ZZ-0002 to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as trippable and is currently at 214 steps.

INITIATING CUE:

The current time is 1010.

This is a time critical JPM. Time will commence when you are provided with the applicable procedures.

When performing calculation, the CRS directs you to use only the tables as directed by SC.RE-ST.ZZ-0002, do not use Figures.

Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Perform SDM calculation and determine SDM is -2441 (+/- 5 pcm) and SAT.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Obtains copy of procedure SC.RE-ST.ZZ-0002(Q). Obtains copy of S2.RE-RA.ZZ-0016 Curve Book.	Provide Rev. 23 of surveillance. Provide Rev. 8 of Curve Book. Evaluator: MARK start time of JPM: _____		
	3.0	Reviews and signs off Precautions and Limitations Section 3.0	Reviews Precautions & Limitations and signs off Steps 3.1 – 3.7.		
	5.1	SELECT the applicable step below (e.g. 5.1.1.A):	Determines 5.1.1.C is the correct step for the current conditions, which is to complete Attachment 3.		
	5.1.1.C	<u>IF</u> SDM is to be verified (per T/S 4.1.1.1.1.a) with reactor critical (Mode 1 or 2) and one or more control rods are inoperable, <u>THEN</u> COMPLETE Attachment 3.	Refers to Attachment 3.		
	Att. 3 2.1	The reactor is in Mode 1 or Mode 2 with $k_{eff} \geq 1.0$	Determines Unit is in Mode 1.		
	Att. 3 3.0	Precautions and Limitations: Section 3.0	Reviews and signs steps 3.1-3.5.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 3 Section 4.1	Section 4 contains the data points required and the calculations to determine shutdown margin.	Determines the following data and calculations: 4.1.1 100% 4.1.2 300 ppm 4.1.3 D @ 227 steps 4.1.4 10,000 EFPH 4.1.5 0 4.1.6 1		
*	Att. 3 Section 4.2		4.2.1 -3841 4.2.2 -3335 4.2.3 728 4.2.4 0 4.2.5 500 4.2.6 0 4.2.7 -5948		
*	Att. 3 Section 4.3		4.3.1 -5948 4.3.2 645 4.3.3 120 4.3.4 2742 4.3.5 -2441		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	4.4	Att 4 Acceptance Criteria 4.4.1 IS THE SDM (Item 4.3.5) EQUAL TO OR MORE NEGATIVE THAN (-)1300 PCM?	Reviews required SDM in MODE 1 or 2 of -1300 pcm and initials 4.4.1. Determines calculated SDM is SAT and initials SAT.		
*			Evaluator: MARK stop time of JPM: _____ when procedures are returned to proctor. Total time from START to STOP is 50 minutes or less.		
			Terminating Cue: Once operator has returned procedures to proctor, terminate JPM.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- M ✓ 1. Task description and number, JPM description and number are identified.
- M ✓ 2. Knowledge and Abilities (K/A) references are included.
- M ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- M ✓ 4. Initial setup conditions are identified.
- M ✓ 5. Initiating and terminating Cues are properly identified.
- M ✓ 6. Task standards identified and verified by SME review.
- M ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- M ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 23 Date 8-22-16
8
- M ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: 8-22-16

SME/Instructor: N. Mulford

Date: 9/29/16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was been declared INOPERABLE at 1000 today due to not moving out during a rod exercise test. You have been directed to perform a Shutdown Margin Calculation IAW SC.RE-ST.ZZ-0002 to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as **trippable** and is currently at 214 steps.

INITIATING CUE:

The current time is 1010.

This is a time critical JPM. Time will commence when you are provided with the applicable procedures.

When performing calculation, the CRS directs you to use only the Tables as directed by SC.RE-ST.ZZ-0002, do not use Figures.